



RX INTERVIEWS

**ACCOUNTING &
VALUATION Q&A**

**RESTRUCTURINGINTERVIEWS.COM
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INTRODUCTION

A Note Before Beginning...

This course was created to teach you things you won't find in any other course out there.

Accounting (and potentially some valuation) technicals will crop up in most RX interviews. These will be extremely similar to the technicals you find in the big, well-known investment banking guides you have probably already read.

The reason why these questions are often asked in RX interviews is because it just demonstrates that the candidate is well prepared. It's assumed that RX interviewees are also interviewing for M&A jobs (because RX jobs are so limited) so candidates should be well versed in classic IB interview questions.

...Because of this, I'm not going to get into creating 500 accounting and valuation questions / answers that aren't overly dissimilar to what you've likely already seen. This course was created to teach you something new, not to rehash what is already out there. So, you should still drill all the classic investment banking questions out there and skim through books like *Investment Banking* by Rosenbaum and Pearl if you have time.

With that said, I am including some of the more common, traditional IB questions used in RX interviews here. Some of these traditional technicals have a RX-specific spin put on them, so I give in-depth answers on how these technical questions apply to a RX job (as often it's quite different than in M&A).

But, to be clear, knowing just these questions alone may not be sufficient. It's annoying to have to study accounting and valuation questions that aren't directly related to the job, I know, but you want to cover all your bases.

The single most important thing – from the traditional technicals – to know is how changes to AR, accrued expenses, deferred revenue, depreciation, etc. flow through the three statements. I've covered some of them here but take the time to make sure you know how they all flow.

Note: Be sure to look at the Bonus 2021 Guide and Bonus 2022 Guide for additional and more advanced technicals that may crop up.

If you have any questions, let me know.

Number of Q&A: 96



ACCOUNTING Q&A

1 Tell me briefly about the three financial statements? How do they (very briefly) link?

You have the income statement (IS), balance sheet (BS), and cash flow statement (CFS).

The income statement reflects the company's revenue, COGS, SG&A, and other tax-effecting items for a *current period of time* (normally either a quarter or a year).

The balance sheet is made up of assets, liabilities, and equity.

The CFS begins with net income – flowing from the bottom of the income statement – then adjusts for non-cash expenses (that do appear on the income statement) in the cash flow from operations section. Then there are the cash flow from investing and financing sections that are rather self-explanatory. Of course, the end cash flows to the top of the balance sheet to complete the connection between the three.

There are obviously more connections than just net income and cash. For instance, you have changes to cash flow from debt issuance or paydown that also must be equally reflected on the balance sheet. Net income also flows from the income statement to the retained earnings of the BS.

1 In distressed situations, which of the three statements do we care about most?

As a general rule you always care most about the CFS, but for distressed companies this is even more true. Ultimately liquidity enables us to figure out what a company can and cannot do to restructure, and a key component of liquidity is cash (remember when I'm referring to liquidity, I'm really referring to the liquidity table that we do beside every cap table).

Note: In Chapter 11 you have to do monthly operating reports showing cash-in and cash-out, which just solidifies how accrual accounting is nice, but ultimately cash is king.

Note: Of course, you can recreate a cash flow statement from the income statement and balance sheet (as long as you have them for two periods) so do be careful with how you word your answer.

1 Philosophically what really is an asset or a liability? Does it differ if the company is distressed?

For a company, an asset is really something that can generate additional cash, or contributes to it in an essential way, in some period(s) in the future. Often the way we think about valuing assets within a company is in some form of discounted cash flows not dissimilar to how we value a company in aggregate via a DCF.



Liabilities are just the opposite. They strip away cash – directly or indirectly – sometime in the future (perhaps over multiple periods).

For a distressed company, something kind of strange happens: assets suddenly are deemed to have less capacity to generate cash – directly or indirectly – in the future, while liabilities strip away roughly the same amount of cash directly or indirectly.

Think about PP&E for a healthy company that then turns distressed – perhaps because their products are no longer in vogue – are these assets worth more or less than before? Obviously less.

Now think about the debt of a healthy company that then turns distressed – perhaps because their products are no longer in vogue – are the liabilities worth more or less than before? Well, from a cash perspective, the same amount is due unless there's a restructuring event of some kind, right? Yes, the market value may be less, but the book value remains the same until that debt is restructured or a bankruptcy occurs.

This can all really be summed up as saying that liabilities are more “sticky” in their cash *drain* than assets are in their cash *gain*.

1 When should something appear on the income statement?

Remember that the IS represents a current period, whatever that period may be, and that near the end of it we have taxes. So obviously what goes into the income statement must affect taxes, otherwise pre-tax income would become distorted by applying the effective tax rate of the company to that added line item!

If you prefer a (bad) analogy, think of a blender. You know you can only put things into a blender that the blades will actually be able to tear up. Likewise, you can't put things on the income statement that don't affect taxes or fall outside of the current period.

1 If we have \$10 in depreciation (a non-cash expense) then how do we treat it on the three statements? Assume a tax rate of 40% throughout.

This is a quintessential banking interview question. You need to know these types of classic questions cold because occasionally you'll be lobbed a softball question like this and no matter how much you know about RX, you really can't get these wrong.

Depreciation applies to the income statement because it affects taxes, and it applies to the current period (it's the depreciation of the thing in the current period). So, if we have \$10 in depreciation expense, then we'll have -\$6 in net income, but are we paying out cash here?

Of course not. So, we add back the \$10 in CFO within the CFS.



Now we have the cash balance (assuming no other changes to the CFS) up by \$4. Does that make sense?

Yes, because without this depreciation we would have paid \$4 more than we otherwise would have in taxes.

On the BS you have assets (cash) up \$4, you have the asset (the thing being depreciated) down \$10, and you have SE (via net income) down \$6.

1 How do we think about capitalizing vs. expensing things? Anything to be mindful of in the context of restructuring?

We capitalize things when their useful life is over a year and then depreciate it or amortize it over the relevant period (by convention).

In distressed, we care a lot about capital leases, because they are considered debt and are often added (if they're large) to cap tables.

Capital leases are generally leases in which the lease runs for:

- At least 75% of the useful life of the asset (most common one)
- Contains a "bargain" purchase price option at end of lease (get the asset at a steep discount)
- There is optional ownership takeover language when the lease ends
- The PV of the lease payments represents over 90% of the asset's fair market value

Note: Don't be confused by the fact that in a Chapter 11 you can assign or reject leases. These can be normal leases for retail space that are *not* necessarily capital leases.

1 Walk me through the distinction between how we handle debt repayments (principal) and interest payments on the IS vs. the CFS?

Interest payments are tax deductible and obviously occur in frequent intervals making it "qualify" as an element to be included in the income statement.

...It will not appear within CFO as, unlike depreciation, you actually have to pay out cash (unless it's PIK interest) and so no adjustments are needed. The IS reflects the cash diminishing already!

Debt repayments are cash, but are they tax deductible? This is why they show up in the cash flow from financing section. Think for a moment about what an income statement would look like if a major piece of debt were all of a sudden paid down? What if a piece of debt was raised?



1 What is deferred revenue? Do distressed companies have more or less of it than average?

Deferred revenue is a liability. It reflects the fact that you've brought in cash for a good or service but haven't yet actually provided that good or rendered that service.

It's a liability until the good or service is provided or rendered, after which it becomes revenue on the IS with deferred revenue being brought down.

Distressed companies will often have more deferred revenue than normal as they'll try to put together deals or sales in which they're promising to deliver something in the future to get cash now.

You see this a lot with TMT companies and sometimes with natural resource companies as well. They'll put together great annual deals and then the quarterly income statements will show deferred revenue being incremented down to reflect the fact they actually provided that service in a given quarter.

1 So deferred revenue can be quite large for a distressed company. What about accounts receivable? What's the difference between them anyway?

The distinction between the two is quite simple, but this is a popular interview question for a reason.

AR (an asset) is simply when a company has made good on providing a good or rendering a service but has not received cash yet. With DR (a liability) cash has come in, but the good hasn't been provided fully yet or the service hasn't been rendered fully yet.

AR is important, especially for distressed companies, because it's usually part of the borrowing base that determines the size of the revolver the company can get (see the other Q&A reports for more on borrowing bases).

AR can be quite large for companies in distress as well, but this is not always an indication of strength. Traditionally you would like to see a company with AR Days under 45; meaning it takes 45 days to turn AR into cash.

If a company is in distress – and those buying from it know it – they may be less likely to actually pay, which will extend AR Days out.

1 What's the difference between noncontrolling interest and investments in equity interests?

NCI means you own between 50-100% of a company. You reflect, within liabilities, the amount of the company you do *not* own.



Investments in equity interests is when you own between 20-50% of a company. This is reflected on the balance sheet, as an asset, for the amount of the company you *do* own.

1 Let's say you own 80% of a company that has generated \$100 of net income. Because you own 80% of it, you have consolidated financials. However, you don't own *all* of it. So how is this \$100 in net income reflected?

On the IS you are reflecting that you own 100% of the company when in reality you own 80% of it, which is obviously not correct. So, you add a line item "Net Income Attributable to Noncontrolling Interests" prior to "net income" and subtract \$20 (20% of the \$100 in net income of the company that you have an 80% stake in).

On the CFS you add back the \$20, because in reality you own a majority of the company and thus have control. So, while the net income is not "yours" the cash is yours "in trust". Therefore, the CFS remains unchanged.

On the balance sheet things will likewise stay the same, we're just reclassifying things. So, there is \$20 in net income not in retained earnings, because we stripped it out before net income on the IS, but we add the \$20 to "Noncontrolling interests" within SE. In other words, even if we didn't strip out the \$20 before net income, and just consolidated the IS all the way through, things would have balanced in the same manner (it just would have been incorrect from a classification point of view).

1 If the company instead is one that you own between 20-50% of, how would you think about having \$100 in income?

When you own this range, you don't consolidate any of your statements, but you need to reflect the fact you've got "income" even if it isn't cash you have discretion over the spending of.

On the IS you create a note or line item below net income called "net income attributable to equity interests" which shows the increase of \$20-50 (depending on % of the company owned by the Parent). You then consolidate that into your actual net income, which will often be called something like "net income attributable to HoldCo" (sometimes "Parent").

On the CFS net income is up by \$20-50, but you subtract that out as you don't actually have cash (you just have net income that could be turned into cash via a dividend by this company you own part of).

On the balance sheet, "Investments in Equity Interests" which is an asset, not part of SE, goes up by \$20-50 while retained earnings (in SE) is up by \$20-50 as a result of "net income attributable to HoldCo" flowing into it.



1 If you hold under 20% of a company, then do you report net income?

No. Sometimes you'll see this if HoldCo used to hold more than 20%, thus it reflected net income from the SubCo previously, but now owns under 20% of it.

In reality, you'll only reflect dividend income or the sale of shares when you hold under 20% because it is treated as a security.

1 What is a common form of interest for distressed companies besides cash? How, if you have \$100 of this, do you reflect it through the statements?

PIK is quite common for distressed companies as it reduced cash interest expense.

Pre-tax income falls by \$100, because PIK is a form of interest. So net income falls by \$60 (assuming a 40% tax rate).

On the cash flow statement, you have cash down \$60, but you add back \$100 since PIK is obviously not cash. So cash is up by \$40.

Cash on the balance sheet is therefore up by \$40. On the liabilities since you have \$100 of new debt (due to the PIK) and within shareholders equity you have retained earnings fall by \$60.

1 Can we have a distressed company with a negative SE?

Sure! If a company is struggling, then it could easily have declining retained earnings to the point where it switches to negative.

If the company has bought back a lot of stock you could also have a large negative treasury stock (just the stock that the company has bought back). Sometimes you'll see companies with quite a bit of cash, but declining equity prices, buy back lots of stock and suddenly find themselves in distress as the business can't be turned around.

1 What is the working capital of a distressed company likely to be?

This is a bit of a trick question. It could be negative, or it could be positive.

For example, a company could have lots of AR (as previously discussed, this isn't always a good thing) and not much DR (as previously discussed, this isn't always a good thing either) and no short-term debt coming due. Therefore, you may very well end up with a very positive working capital number making the company seem quite stable. However, there could be massive maturity walls coming due around the corner that mean a restructuring is all but certain.



1 Do you think we use operating working capital in RX a lot?

You never want to give definitive answers to these kinds of questions, but generally it's not going to be overly informative to use operating working capital.

OWC = current assets excluding cash & investments – current liabilities w/o debt

Well cash is king in distress and debt is incredibly important (because there's normally an awful lot of it!) so how informative would this really be to us? It would tell us that maybe the company, if it had a right sized balance sheet, would be healthy. But it wouldn't tell us much about whether the company is distressed in the first place or how the capital structure should be reconfigured.

1 We use EBITDA a lot in RX (for leverage ratios, etc.) but does it really tell us anything about the health of the company? What doesn't get captured?

EBITDA itself is not always overly reflective of the health of a company. Capex is not captured, one-time expenses are not captured, and maturity walls that are imminent and can't be refinanced are not captured.

However, in investment banking we have to have some agreed upon ways of thinking about companies. Even GAAP itself is imperfect at capturing the health of many companies.

EBITDA may be imperfect, but it is the convention. Further, most of what we look at is EBITDA in the context of a ratio. (Debt/EBITDA) and (EBITDA/cash interest) are still useful indicators of the capacity for the company to meet its debt funding obligations and important to the covenants that undergird that debt.

1 What do you normally add back to EBITDA for distressed companies to get an Adjusted EBITDA?

- Bad debt expenses
- Asset write-downs
- Goodwill impairments
- Legal fees
- Restructuring fees

1 Are deferred tax assets and deferred tax liabilities common for distressed companies. What are they really? What is more common to see?

A DTA simply means that you are paying too much tax now and as a result will pay less in the future. Normally originating from net operating losses (NOLs) that come from negative pre-tax income. NOLs, that result in DTAs, are an incredibly important asset for a company, especially one coming out of Chapter 11.



However, not all companies can retain their NOLs. Please see the RX guide for more on how NOLs are treated in bankruptcy as it pertains to retaining their value and not having it wiped out.

A DTL is just the opposite: you paid too little taxes today, so will have to pay more cash taxes tomorrow.

For complex companies, or companies with volatility in earnings, you'll see DTLs and DTAs sitting on the balance sheet. This reflects the fact that the DTLs and DTAs originated in different manners and at different times.

1 Speaking of NOLs, how do they intersect with DTAs? Are they two sides of the same coin?

More or less, yes. NOLs are represented through DTAs. Or another way to think about it is that NOLs are the precursor to DTAs. When NOLs are utilized the DTA will change by $(\text{NOLs new or existing} - \text{NOLs used}) * (\text{Tax \%})$.

So, for example, if you have \$500 in pre-tax income, you'll have \$200 in income taxes. If you have \$100 NOL you will be able to offset cash taxes by \$40, so you really only have \$160 in income taxes to pay in cash.

It's important to remember that NOLs are *not* how much you can reduce cash taxes by, it's how much can be multiplied by the prevailing tax rate of the company to reduce cash taxes by.

1 Let's say there are \$100 in NOLs on the books – from being in a distressed situation – but they just pull in \$100 in pre-tax income. What happens?

NOLs are a conduit so we proceed with the IS as normal.

\$100 pre-tax income leads to \$60 net income. Meaning \$40 in taxes were "paid".

The *actual* cash taxes paid reflect the use of NOLs. \$100 in NOLs * 40% tax rate means that \$40 is eligible to be applied against the \$40 in taxes that must be paid.

On the cash flow statement, you have \$60 in cash (due to net income) then you add \$40 since the DTA (which can be thought of as housing the NOLs) decreased (remember when assets decreases, that's a source of cash for the company). So, you have a \$100 increase on the cash flow statement.

On the BS you have cash up by \$100, you have the DTA down by \$40, and you have \$60 in retained earnings as a result of net income.



1 Walk me through the three statements when accrued expenses increases by \$100?

Remember that accrued expenses are expenses that have been accrued, but haven't yet been paid.

So pre-tax income goes down by \$100, net income goes down by \$60. Cash on the CFO starts off down by \$60, but since accrued expenses are non-cash, you add back \$100 so you have CFO up by \$40 (no other changes to the CFS are made).

Remember that accrued expenses, like all other non-cash expenses added back to the CFS, are being adjusted to reflect the tax "savings" that have occurred.

On the balance sheet cash is up by \$40, liabilities are up by \$100 (since accrued expenses are a liability), and equity is down by \$60 (as retained earnings falls due to the -\$60 net income).

Note: When accrued expenses are then paid in cash nothing happens on the income statement as these expenses occurred in a *previous* time frame. So, cash is down on the CFO by \$100, which flows to assets on the balance sheet being down \$100 and liabilities being down \$100 (due to accrued expenses going down \$100).

1 Generally speaking, in the CFO when you have an increase in assets or liabilities is that a gain in cash or loss?

Assets are losses in cash and liabilities are generally gains in cash. Think about an asset as something you're paying cash for now that you hope generates you more cash over the useful life of the asset purchased.

1 When inventory is purchased for \$100, what happens to the three statements?

This is a common question to gauge whether you understand what hits the income statement or not. Nothing happens to the income statement because the inventory hasn't been sold, it's just been purchased. CFO is down \$10 because you've used cash to buy the inventory and this flows to the asset side of the balance sheet where cash is down \$10, inventory (an asset) is up \$10 and nothing else occurs. Because asset line items go up by \$10 and down by \$10 the balance sheet isn't out of balance.

Note: When inventory is sold then it turns into revenue and of course hits the IS.

1 Let's say a company sells an asset for \$200 that had a book value of \$100. What happens?

On the income statement you record the gain of \$100, which is reflected in a \$60 increase in net income. CFO is up by \$60, but you net out the entire gain (as it will be reclassified



in the CFI), so CFO is down by \$40. CFI is up by \$200 (the sale) so the CFS, in totality, is up by \$160.

On the balance sheet you have cash up by \$160, an asset off the books (\$100) so assets are up by \$60. You have no change in liabilities, but the \$60 in net income flows through to retained earnings so SE is up by \$60. Therefore, everything balances.

1 Asset write downs are more common than asset gains for distressed companies. So, what happens when you have an asset write-down of \$100?

Asset write downs affect the income statement since losses operate as a tax shield.

So, the income statement, pre-tax, is *down* \$100, and after tax down \$60.

Within CFO it begins with -\$60, but an asset write-down is non-cash, so we add back \$100 making CFO up by \$40.

On the balance sheet cash is up \$40, but the actual asset in question (say PP&E) is down by \$100, so the asset side of the BS is down \$60. Nothing occurs in liabilities, but the SE is obviously down by \$60 as net income flows into retained earnings.

1 Deferred taxes are quite common in distressed companies. Let's say that you have an income tax expense of \$100, but \$50 is paid in cash and \$50 is paid sometime in the future. What happens here?

The IS stays the same as it doesn't matter whether something is cash or non-cash.

On the cash flow statement remember that when you hear CFO you should think: "are we dealing with something non-cash here?" Obviously \$50 is non-cash here so we add back \$50 here.

On the balance sheet cash will be up by \$50 (from the add back within the CFO), but we now have a liability of \$50 for deferred taxes (reflecting the fact that the company will have to pay \$50 in cash at some point for these taxes).

1 Asset write downs are common in distressed companies, but so are debt write downs if a company is *really* distressed. Let's say we have \$100 in a debt write down. What happens?

Remember when we discussed asset write downs, net income *fell*. When debt is written down net income *increases*.

This should make sense once thought about. When an asset is written down it reflects that an asset and the string of cash flows, literally or metaphorically, attached to it have diminished. When debt is written down it reflects the fact that cash flow, that literally would have been directed toward paying that debt, has been freed up.



So, the pre-tax income goes up by \$100, net income goes up by \$60. On the CFS we have cash up by \$60, but we need to reflect the fact this is a non-cash expense, so we subtract \$100 to get to -\$40 in CFO.

On the balance sheet we have cash down \$40, we have liabilities (debt) down \$100, and we have SE (via retained earnings) up by \$60. Thus, we have balance.

1 So, we have debt write downs. What about just paying the debt of say \$100 (not interest, but principal)?

Nothing happens to the IS as we aren't affecting taxes directly by paying down debt. Think about what net income would look like if it's normally \$4mm, but we're paying back \$400mm in debt (or refinancing it).

On the CFS, under cash flow from financing (CFF), we decrease it by \$100 to reflect paying down the debt. This reduces cash, all things considered, by \$100.

On the balance sheet we have cash down (asset) of \$100, but we have debt down by \$100.

1 How do you calculate trailing twelve months (TTM) EBITDA? Do we use that?

For companies that have cyclical earnings, then yes it can be used. However, in distress EBITDA tends to fluctuate a lot because of one-time adjustments, outright decline in revenue, etc. so we tend to just focus on what the trend of EBITDA is.

Trailing twelve months is just = new period + twelve-month period – old partial period

1 Let's say we are selling a widget for \$100, and the cost is \$50. What happens when a sale occurs (in cash)?

On the IS you have \$100 in revenue, COGS of \$50, and therefore pre-tax profit of \$50 (controlling for everything else). Then you have \$30 in net income.

On the CFS you have \$30 in cash, inventory has decreased by \$50 (which is a cash addition; assets down, cash up), so you have \$80 in cash for the CFS.

On the balance sheet, cash is up \$80, but you have inventory down \$50. So, you have \$30 up on the asset side. And on the SE side you have \$30 reflecting the retained earnings increase from net income going up \$30.



VALUATION Q&A

1 What are the three major valuation methodologies?

Comparables (public comps), precedent, and DCF.

In RX, comps and precedent aren't used remotely in the same way. You'll have precedent deal *structures*. But remember what you're really doing is working within the confines of a capital structure specific to a given company and trying to right-size it.

Some deals will look similar - by virtue of having similar capital structures, etc. – but there's no football field analysis like in M&A.

Remember that in M&A what we're doing is trying to come up with the right range of valuation for a company. In RX, we're saying, "this company needs to be right-sized, or it'll go bankrupt" or "this company needs to go bankrupt (Chapter 11) to get right-sized" so discussions around the real valuation of the company are per se unimportant relative to M&A.

1 Would a DCF or a liquidation valuation give a better valuation for a distressed company?

Like anything, it will depend.

A liquidation valuation sets a hard floor on the value of a company. What we're saying is if we could just liquidate all the assets of a company, how much cash would we get and how would that cover creditors down the capital structure. Liquidation, of course, is the worst-case scenario and involves a U.S. Trustee supervising the process.

A DCF will give a higher valuation if the company is not overly distressed. However, remember that much of the value of a DCF comes from its terminal value (5 years to infinity) and for a truly distressed company perhaps they will be bankrupt within five years or sooner.

So, for a distressed company that can almost certainly get right-sized out-of-court, a DCF will produce a higher valuation. For a distressed company that almost certainly will go through an in-court procedure, a liquidation valuation may be closer to reality.

1 Is EBITDA analogous to cash flow?

It's a proxy, but it excludes certain things that are important like capex, one-time fees (high for a distressed company), debt repayments, and interest.



1 How do we calculate levered FCF and unlevered FCF? For distressed companies, what matters more?

For unlevered FCF, it'll be $(EBIT \times (1-T) + \text{non-cash items} - \text{changes in operating assets} / \text{liabilities} - \text{capex})$

For levered FCF, it'll be $(\text{net income} + \text{non-cash items} - \text{changes in operating assets} / \text{liabilities} - \text{capex} - \text{mandatory repayments})$

For a distressed company, what you really want to see is how a company looks without having any kind of interest expense and principal repayments to ascertain what kind of capital structure the company can actually bear.

Unlevered free cash flow is associated with enterprise value as it doesn't include interest and debt repayments, meaning it is cash flow available to everyone. In distress you care more about this number as we are dealing with debt holders, and we care about how they're getting paid to hold existing/new debt.

Levered free cash flow would be associated with equity value as levered free cash flow assumes interest and debt repayments have been made (thus why net income is used, and mandatory repayments are in the equation). Obviously for a truly distressed company levered free cash flow will likely be quite small given the mounds of debt in the capital structure the company can barely maintain (if it can at all).

1 Why would it not make sense to use equity value / EBITDA? Would this number, even if it shouldn't be used, be high or low for a distressed company?

Because EBITDA belongs to the entire capital structure – including debt holders – so you can't use equity value, you have to use enterprise value (which would take into account all those with interests in the company).

It'll be low for a distressed company, perhaps below 1, because equity holders have the lowest "interest" in the company so as a result equity value declines severely when a public company nears distress. EBITDA can still be quite stable, but if it appears a company will need to restructure, then equity will plummet (as it has the lowest possible "claim").

1 Would you expect EBITDA to be higher or lower with large lease expenses vs. outright ownership?

This is a common style of question. For RX interviews, it's worth drawing the distinction between capital leases and operating leases noting that capital leases are effectively debt not expenses.

With that said, EBITDA will be higher when you own vs. lease because DA is tacked onto the end. Whereas lease expenses are categorized within general operating expenses.



1 Will EPS and P/E be higher for a public company in distress vs. a healthy company (say comparables that are part of the S&P 500)?

Don't be so quick to answer this question. It could go either way for the P/E ratio.

On the one hand, if the company has had diminished earnings (but the market believes there is a viable turnaround plan) relative to healthy comparables then the price could be more "sticky" making the P/E ratio higher. However, earnings could also not decline too much but the capital structure could be so onerous that equity declines (reflecting the belief a Chapter 11 will happen at some point) making the ratio very low.

EPS is almost always lower as you are going to have roughly the same denominator (avg. shares outstanding) as distressed companies don't have the cash for massive buyback programs or anything. So, a distressed company almost invariably will have lower and declining earnings relative to a healthy comparable and thus a lower EPS.

1 Can sum-of-parts valuations be used for a distressed company?

Absolutely. If a company has quite a few business lines, it's entirely possible that the capital structure assumes all business lines are doing well, when in reality only two or three out of five (for example) are.

For a large enough company, you may break out business lines and assign EBITDA multiples in line with comparables for the healthy line items, and much lower ones for the unhealthy business lines, to come up with an EV.

This is normally done when you're advising a company on asset sales prior to bankruptcy; to bring cash in to be used to help restructure the capital structure that remains after the sale.

Note: when asset sales are involved, normally a M&A team will join the RX group and do the modelling on the deal.

1 Is equity value relevant to distressed companies?

Sure. Shareholders are part of the capital structure, they just generally come into a restructuring event with very low value and almost never get anything out of a Chapter 11 process.

Equity value is also important insofar as it informs enterprise value, which is something we care about as it represents the value of the company in aggregate.

Note: You may always want to caveat Chapter 11 discussions regarding equity by saying that you will sometimes give a "tip" to equity holders (normally just a few percent of reorganized equity).



1 What is the enterprise value formula?

$$EV = \text{equity value} + \text{debt (including capital leases)} + \text{preferred stock} + \text{non-controlling interests} - \text{cash} - \text{NOLs} - \text{securities \& equity investments}$$

Note: NCI is added because you'll have the financial statements consolidated if you own over 50% of the company.

Note: We also add in capital leases to debt, because they are functionally a debt-like obligation and that's why we include it in the cap table as well.

Note: NOLs are taken off because they allow for a cash savings sometime in the future via reduced taxes. NOLs are very important to distressed companies; they represent a meaningful asset.

Note: Only take off equity interests (equity investments, where you own 20-50%) if the financials do not include net income attributable to equity interests, as previously discussed.

1 We use diluted equity in the EV equation. Is that *really* necessary for RX? Are there dilutive effects really occurring when the stock price of a distressed company probably is falling?

This is a great question to gauge if you understand how a company that may need to be restructured will differ from a healthy company.

You always use diluted equity value in the EV equation to account for options, converts, and preferred stock.

While it's easy to assume that options won't be in-the-money (ITM) so you don't need to worry about them, options (for management, in particular) can often have very low strike prices so could very well vest and be turned into more equity.

Likewise, convertible debt has been very popular over the past decade. While it's unlikely that convertible debt is ITM, it could be the case and is very important since if it does convert then you obviously have a different debt load.

To find the dilutive effect you use the treasury stock method for options. For converts, it's obviously pre-ordained as to what the dilutive effect will be, so you look to the terms of issuance. If the converts are OTM, then they're just added to debt.

1 What's the diluted equity value if you have 1,000 shares outstanding, a share price of \$5 and 100 options struck at \$3?

Equity value prior to adjustment will just be shares multiplied by the current share price, or \$5,000.



In order to utilize the option, the holder will need to buy at \$3 and it is assumed the company uses the cash gained from this to buy back stock at market prices. So, the company has \$300 in cash that it uses to buy back 60 shares.

That means the fully diluted amount of shares outstanding are 1040 ($1000 + 100 - 60$), giving an equity value of \$5,200 (\$200 more than previous).

1 What are the steps to finding the dilutive effects of convertibles?

1. Find out how many convertible bonds you have
 - a. Take the nominal amount of converts divided by the par value (\$1,000)
2. Find out how many shares you get per bond
 - a. Par value (\$1,000) divided by convert price
3. Find the total new shares
 - a. Shares per bond (found in #2) multiplied by number of convertible bonds (found in #1)
4. Add total new shares to pre-existing share count to get total diluted number of shares
5. Multiple total diluted number of shares by the prevailing market price to get the fully diluted equity value

1 Philosophically, why do we even care about enterprise value (EV) for a distressed company?

In restructuring we care about two things: what a company is worth, and the debt load the company can handle (from a liquidity perspective).

At the basest level, we care about EV because it informs whether a company *should* be restructured or simply liquidated. That's why, in a Chapter 11, the court uses a fairness test to determine if it's even worthwhile to have a company restructure or if it should be liquidated (Chapter 7) and just pay back creditors by absolute priority.

1 Is adding back the book value of debt *really* giving a fair reading of EV?

This is a great question, because in M&A you add back debt because the debt will be repaid in an acquisition. So using book value makes perfect sense.

For a company that is likely to restructure, is this relevant? By definition we're going to be changing up the terms of debt, probably trying to get rid of some of it in some way, so why use the book value pre restructuring? Also, the market price is likely at a heavy discount to the book value.

There's no obvious answer here. The reality is, as is often the case in finance, you use what you're given and try not to rely on too many hypotheticals.



We show in cap tables what market prices are, what the market price times the debt is (e.g. \$100mm * 0.78, if the debt is trading at \$78), and then calculate leverage ratios based on that. This is meant to prompt the question: what if we are able to convince holders of debt to exchange their current debt (on the balance sheet at book value) for an equivalent amount at market prices (thus heavily reducing the principal amount of debt).

So, in RX where possible we do use market prices and do EV calculations based on market pricing and on book value. However, the market pricing is meant to inform the above hypothetical and we don't necessarily think, "this is what the company is really worth" as obviously the company really does have the *book value* of debt until it's restructured.

Also, restructurings don't always end up with less face value debt. Sometimes in order to diminish cash interest you'll be adding PIK (more debt) or exchanging bonds for a slightly larger issuance at a lower interest rate (more principal), etc.

The reason why this is a great question is that there is no true answer. The answer forces you to show that you understand that how we think about debt in RX is much more malleable than in M&A. It's not just a number in an equation.

1 What is a DCF really doing? How does it apply to RX?

A DCF is fundamentally about taking the free cash flow of a company over its entire duration and then saying, "What is that worth today?"

Given that, to get to today's value you have to discount these cash flows. So, the near-term cash flows matter much more than the far away cash flow.

So even though, philosophically, we're saying this company goes on into perpetuity (that's what the terminal value represents in an abstract sense) those far out cash flows accrue very little to the actual enterprise value that the DCF gives you.

In RX, a DCF matters in so far as it gives you an enterprise value in a different way. However, given how by definition a company likely to be restructured will have volatile FCF, it's hard to rely on this for much.

It's much more common to see a liquidity roll forward (which uses FCF) in a pitch or live deal than a full-blown DCF.

1 What are the two components of a DCF?

The first, is the near-term projection which can be made with a bit more precision. Here you project out the company's financials, getting to (unlevered) FCF each year, for normally five years (although sometimes longer). You then discount each of these years (using mid-year convention) using the WACC (weighted average cost of capital).



The second part is the terminal value, which can be thought of as what the company's value is, today, from where the near-term projection ends out to infinity. Almost always a multiple will be applied against the year five (or wherever the near-term projection ends) EBITDA. You then discount this value back using the WACC as well.

You then add the first and second piece together to get enterprise value.

1 How do we get unlevered FCF?

- Start at either revenue (then take off COGS and op. expenses) or just start at EBIT
- Multiply EBIT by $(1-t)$ to adjust for taxes
- Add back D&A and all non-cash expenses (since these don't truly draw cash)
- Account for changes in operating assets/liabilities (can be positive or negative), if negative that means assets > liabilities
- Take out cap-ex

1 How about levered FCF?

- Start at either revenue (then take off COGS and op. expenses) or start at EBIT
- Subtract interest expense (and add interest income)
- Multiply this number by $(1-t)$ to adjust for taxes
- Add back D&A and all non-cash expenses (since these don't truly draw cash)
- Account for changes in operating assets/liabilities (can be positive or negative), if negative that means assets > liabilities
- Take out cap-ex
- Take out mandatory debt repayments

Note: For distressed companies, mandatory debt repayments will often make levered FCF an almost nonsensical number over the short term. This is, among reasons already discussed, why unlevered FCF is used.

Note: Remember that levered FCF applies only to equity value; it's essentially the cash available to shareholders after adjusting for what is owed to the rest of the capital structure (that is more senior). In RX it's by definition unlikely to be much!

1 Would you expect a bigger difference in levered vs. unlevered FCF for a healthy company or an unhealthy company?

The difference will almost certainly be much larger for an unhealthy company.

Think about it this way: levered FCF "belongs" to shareholders. If a company is distressed, it means there's pressure somewhere in the debt-side of the capital structure (senior to equity, of course). So, if things are rocky up higher in the capital structure, things must per se look much, much worse for equity (meaning there must be very little FCF left for equity, which is what levered FCF is telling you).



In a healthy company, with a capital structure that is not onerous, there may be tons of cash left over for equity holders because it doesn't have onerous levels of debt so relatively speaking the spread between levered vs. unlevered will be less large.

1 What is (unlevered) FCF really telling us?

We're really saying, after adjusting for things that are non-recurring, what is our actual cash position that the company can put to work in some way.

In RX, more than in any other area of finance, cash is very important. It informs the way MDs think about what options really exist for a company out-of-court.

This is why we do things like liquidity roll forwards that use FCF to find out the liquidity available to the company moving forward.

1 What's the formula for WACC? How do we determine costs for a distressed company?

$$\text{WACC} = \text{Cost of equity} * (\% \text{ equity}) + \text{cost of debt} * (\% \text{ debt}) * (1 - t) + \text{cost of preferred} * (\% \text{ preferred})$$

The percent refers to, for example, the % of equity relative to the entire capital structure.

For a distressed company, things get tricky. Often you'll use the CAPM model for determining the cost of equity, with a large distressed premium.

For cost of debt, we normally take prevailing yields and depreciate them if they're highly distressed (due to the fact the yields would bounce with a proper restructuring done, so prevailing yields could arguably be too high). Preferred shares are treated the same way if they are in the capital structure.

1 What's the CAPM model?

The CAPM model is used for determining the cost of equity. The cost of equity is therefore: $\text{CoE} = \text{risk-free rate} + \text{equity risk premium} * \text{levered beta} + \text{distressed premium}$

The risk-free rate almost always refers to the ten-year treasury rate. Beta refers to the risk of comparable companies (you want to narrow the universe away from non-distressed companies, if possible) and equity risk premium is a set number you'll pull (can find on Bloomberg).

The distressed premium refers to some percent that is bolted onto the end to bump up the cost of equity. This is done to reflect the fact the distressed nature of the company and that the beta is probably going to be predicated on a set of companies that are much healthier (so the cost of equity is artificially low).



1 Does cost of equity even make sense for a distressed company? What is it really telling you?

Cost of equity isn't telling us what the cost of raising new debt is, it's telling you what an equity investor could expect to earn.

For a distressed company, things get much more binary. Equity could either go to zero (in a heavy restructuring resulting in a Chapter 11) or a turnaround could occur from already distressed equity values and thus result in large equity gains!

So, the notion that a distressed premium would be added makes directional sense. It reflects the fact that there are potentially outsized gains from holding equity.

1 How do we get to the beta we use in the CAPM model?

There are really two steps here:

1. Selecting comparable companies
2. Unlevering those betas (as they will be levered to their own capital structures) and then re-levering them based on the actual company's capital structure

The first thing that obviously needs to be considered is what are truly comparable companies. We should pick those in a similar industry and that are not doing overly well (if possible) to get as close to our company as possible.

Of course, you aren't going to have a broad universe of comparable companies that are distressed. So, if there aren't more than four or five, just use healthy companies (that's what the distressed premium is for).

Then you pull their levered beta, from the Bloomberg terminal your group will have access to, unlever them all (using the formula below), and choose the average (if the range is small) or median (if range is large) from this set.

You then use the re-lever this average or median value. This levered beta is what you ultimately use in the CAPM formula.

Unlevered Beta = $LB / (1 + (1 - t) * (\text{debt} / \text{equity}))$

Levered Beta = $ULB * (1 + (1 - t) * (\text{debt} / \text{equity}))$

As you can tell from the formulas, the whole point of unlevering and then levering the beta is to adjust the levered beta you end up using to your specific capital structure.



1 Would you expect distressed companies to have positive betas or negative betas?

This is a great question to ask that gets a bit beyond just asking the interviewee what beta is (which almost all will know by heart).

A negative beta means that the equity moves counter to the market in aggregate. So *almost* all companies are going to have positive betas as a high tide lifts all boats.

However, for a distressed company, this won't necessarily be true. Very few distressed companies, that will actually need to be restructured, will have their stock price moving up significantly even if we're in the midst of a bull market.

So, in a bull market you would expect negative betas. In a bear market you would expect very high positive betas. Why? Because if the market is going down, distressed companies will be going down even more than the aggregate market (as they are the weakest companies).

1 For a distressed company – with bonds trading down to say even 75 – will cost of debt or cost of equity be greater?

It may be tempting to say the cost of equity will be less than the cost of debt if bonds are trading down enough. However, when in doubt just remember the capital structure is a waterfall.

You begin with debt; you end with equity. No matter how much the cost of debt may be the cost of equity per se must be more as it is even further down the capital structure and thus buying equity is even more risky / volatile than buying debt.

1 What will a DCF of a distressed company end up looking like?

The short answer is that it can look "off" in a number of different ways.

Traditionally, the terminal value will make up a big chunk of the enterprise value of the company. For a distressed company, does this really make sense though? Well if the company has negative FCF over the next few years of projections then the enterprise value per se will be entirely composed of the terminal value.

...But for a truly distressed company, is it reasonable that the company will even be around – in at least its current state – before year 5 or wherever the near term of your DCF ends? Perhaps not.

Likewise, you can apply a very low multiple to the far-term segment of your DCF, to account for the distress, but then it's a question of having an enterprise value made up heavily of discounted FCF over a short segment of time.



It's important to remember that traditional M&A valuation methods – that you likely know very well already, at least theoretically – are used for a singular goal: finding the fair value (or range of fair values, to be more precise) that a company should spend for another company.

In RX the singular goal is right sizing a company's capital structure to ensure it can be an ongoing concern.

Part of that requires understanding if the company can be an ongoing concern, which means having a general idea of its value and long-term viability. But ultimately RX deals with the realm of the immediate (rectifying maturity walls, high cash interest, etc.) whereas M&A deals with ensuring that the long-term value of the company purchased aligns with the value paid. These are very different aims and involve very different approaches and tools to be utilized.

1 Will a company with more debt have a higher or lower levered beta? What does this lead to?

Higher. It leads to a higher cost of equity in the WACC and thus a higher discount rate and lower EV (all else being the same).

1 What single changes to a DCF model will produce the largest change?

Changing the WACC will because it underpins everything (applying to the near-term, by discounting the FCFs and the long-term by discounting the terminal value).

1 Why is mid-year convention utilized with DCFs?

To represent the fact that you don't get cash as a stub at the end of each year. Mid-year simply means using: 0.5 for the first year, 1.5 for the second year, etc.

1 What is the philosophical flaw with a DCF that precludes its use for seriously distressed companies?

At the end of the day, a DCF is predicated on Unlevered FCF and EBITDA (via the near-term period, and far-term period, respectively). Both of these formulas try to regularize finances, which is great for traditional companies. You don't want non-recurring items to affect them too much and distort the true, long-term financial picture of the company.

However, in distress you're dealing with near term disruptions (due to the capital structure in conjunction with the company not having great growth on a levered basis) that make it hard to envision the company being a stable, ongoing concern for years into the future.

That's why we care much more about things like liquidity roll-forwards and just what EBITA, FCF, etc. are *right now*.



M&A is focused on ensuring that the valuation the deal is completed at is viewed as being reasonable and fair *in the future*. This is hard to do and requires a triangulation of models in order to find estimates of what FCF, EBITDA, etc. could end up being years down the road in order to produce a reasonable, fair EV.

However, with distress we aren't thinking about years down the road. We're thinking about right now. We're thinking about how we can right size this company *right now* (or at least within the next year or so) that will ensure that EBITDA, FCF, etc. are properly aligned with the debt load of the company.